

Qatar-4b

R-2348

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_231105 · created 2026-08-19T01:02:15+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460253.7447 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.019
Transit depth (Rp/Rs)^2	1.94 +/- 0.53 [%]
Orbital Inclination (inc)	85.46 +/- 1.56
Airmass coefficient 1 (a1)	1.002 +/- 0.018
Airmass coefficient 2 (a2)	-0.001 +/- 0.047
Scatter in the residuals of the lightcurve fit is	1.82 %
Best Comparison Star	#3 - [131, 186]
Optimal Aperture	3.34
Optimal Annulus	12.51
Transit Duration (day)	0.078 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.139 $\pm$ 0.019 vs 0.138 (deviation 0.05 σ)
Duration fit vs published	0.078 d vs 0.0892 d (deviation 1.01 σ)
Mid-time O-C	-4.2 min
Depth SNR	7.3
Residual scatter	1.82 %

Light curve

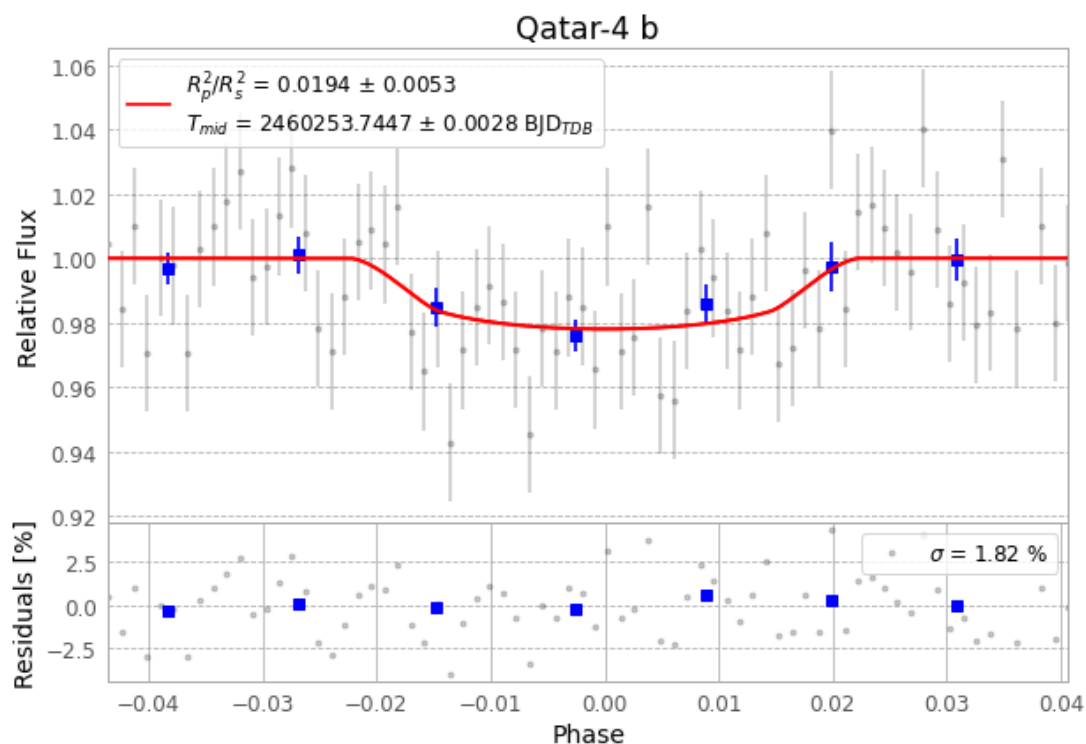


Figure 1 — FinalLightCurve_Qatar-4 b_2023-11-04.png (SHA-256 e2d1265988b31cfc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [274, 196], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9df619832ab214c1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

74 FITS frames (49 MB), Qatar-4231105035115.FITS → Qatar-4231105073015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-231105.log (708b829bcda6...), FinalLightCurve_Qatar-4 b_2023-11-04.png (e2d1265988b3...), FinalLightCurve_Qatar-4 b_2023-11-04.csv (ae5ce3f6fd5a...)

Compute resources

Wall time 112.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 01:02Z.